An Exploration of the Suitability of Cement Substitutes in Stabilised Rammed Earth.

IB Design and Technology Extended Essay

RESEARCH QUESTION

With reference to structural integrity and environmental consideration, what would be the most suitable cement alternatives for the construction of *stabilised* rammed earth structures?

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1 Introduction

Earth walling is recognised as one of the oldest human building methods, dating back to the earliest settled civilisations along the Mesopotamian Basin. These simple earth walls were not structural apart from standing and affording decent protection from wind and sun. Roof structures were typically light timber beams covered with straw and mud. The buildings would last approximately for two generations or so depending on the local climate. It existed for centuries as a useful and, though not recognised at the time, an extremely low embodied energy building method.

Modern "stabilised" rammed earth (SRE) emerged only recently, since the 1970s, and is used as a structural walling system. SRE walls can carry roof loads and point loads such as veranda beams or cantilevering loads. SRE is used by architects as a thermal mass panel for both exterior facades and as internal walling. It is admired for its natural sedimentary appeal.

The modern SRE wall industry remains a specialist/niche product in the current building market. The pioneering and development of stabilising the walls using small portions of cement has seen it gain popularity and used for thousands of both domestic and commercial buildings. The market for SRE is gradually increasing as new variations emerge including precast panelling for high rise facades and ultra-thin panelling for interiors.

Rammed earth is seen as a sustainable construction substitute to solid brick/concrete houses and commercial buildings as the embodied energy of rammed earth wall is exceptionally low compared to say a sold brick wall which is 3.5MJ/kg *(Crawford)*.

Considering the processes involved in manufacturing a fired clay brick, it requires mining the clay, puddling the clay, forming the clay into a moulded shape, baking the bricks at extremely high temperatures for extended periods, stacking and sorting the bricks, then transporting

them to a retailer, then later to the site. Lastly, they need to be laid using cement stabilised mortar.

SRE however has comparably few processes. Local gravel is excavated from a hillside, screened and stockpiled. This gravel is transported typically less than fifty kilometres to the construction site, blended on site and compacted into formwork. With reference to the life cycle analysis (LCA), this depicts the disparity between rammed earth and brick.

The machinery used on a SRE building site uses an approximate 25 litres per day for a skid steer loader and an air compressor. Additionally, the walls are load bearing, so have no requirement for additional steel or timber frames.

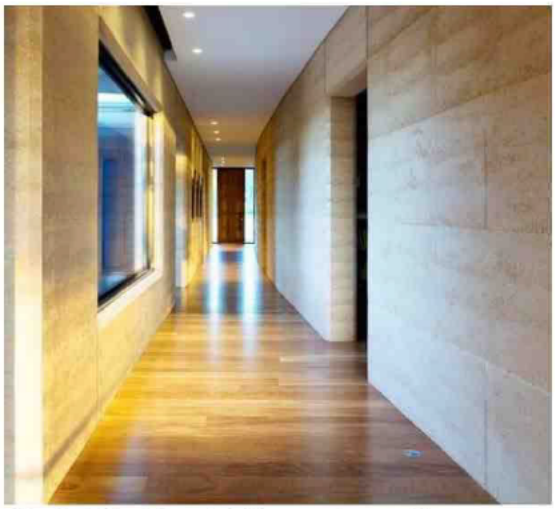


Fig 1.1 Residential Home, Australia
(Earth Structures Group)



Fig 1.2 Bushey Cemetery, UK
(Earth Structures Group)

Currently, SRE uses an average of 8-10% cement in the walls which allows for the strength of the buildings to be structurally comparable to concrete and brick in terms of compressive and tensile strength and durability.

However, the addition of cement, which has a high embodied energy, is recognised as compromising the sustainable credentials of SRE *(Downton)*. With increasing concerns of environmental impacts that comes with the cement production, including increased demand

for more sustainable construction industry, environmental and social demands have subsequently led SRE contractors to look for alternative stabilisers to cement.

On the other hand, traditional "un-stabilised" rammed earth walls have a limited lifespan in terms of shrinkage cracking, water resistance, compressive and tensile strength and surface durability, modern "cement stabilised" rammed earth walls have a virtually unlimited lifespan on all these fronts. Un-stabilised rammed earth that portrays similar structural benefits to that of SRE requires a full portal frame made of steel or timber to acquire this level of durability. Provisioning these steel or timber frames outweigh the embodied energy of the cement used.

This essay will focus on the research for alternative stabilisers in SRE, to improve the environmental footprint of the modern earth building industry on the global climate crisis. Dematerialisation of SRE will mean a change in material without affecting the functionality and integrity of the product. This way the durability and structural appeal will remain alongside the sustainable improvement of modern stabilised rammed earth. If a conclusion is found, this could open countless possibilities for cement free or cement-reduced construction materials, and the progression toward more environmentally conscious buildings. Leading to question: *With reference to structural integrity and environmental consideration, what would be the most suitable alternatives to cement as a stabiliser for the construction of stabilised rammed earth structures?*

2 Initial Background Research

2.1 The construction of stabilised rammed earth

Stabilised rammed earth walls are typically the product of a combination of blended aggregates, which is recognised as road base material, typically 18mm minus in size. Road base material must include small portions of clay, minimising the risk of shrinkage (cracking) combined with even spread of mineral particle sizes, maximising the density of the compacted material. The addition of Portland grade cement, a small amount of fresh water and a silane/siloxane water repelling admixture is added to achieve an optimally dense, damp, compactable consistency.

Built as a series of independent, monolithic panels, this freshly mixed compound is placed between specialised, reusable, recycled thermoplastic polymer (polypropylene, polyethylene) formwork. The formwork is spaced 300mm or 450mm apart, and the compound is rammed in 150mm deep courses using pneumatic rammers. The process of ramming courses is repeated until the full height of the panel is completed. Typically, there is no steel reinforcing (unlike concrete), and roof loads are held down using cast-in bolts.
The following day the formwork is stripped revealing a massive masonry wall. The shape of the building is constructed using a series of these interlinked SRE panels (Figure 2.1.1).

The appearance of each wall is unique, with an off-form sedimentary finish and a natural colour and texture being the defining feature of rammed earth. Depending on the quarry aggregate from where the base material was derived, each building's walls vary in aesthetics. Local quarries are chosen as the short delivery of material limits fuel consumption and subsequently, total emissions.

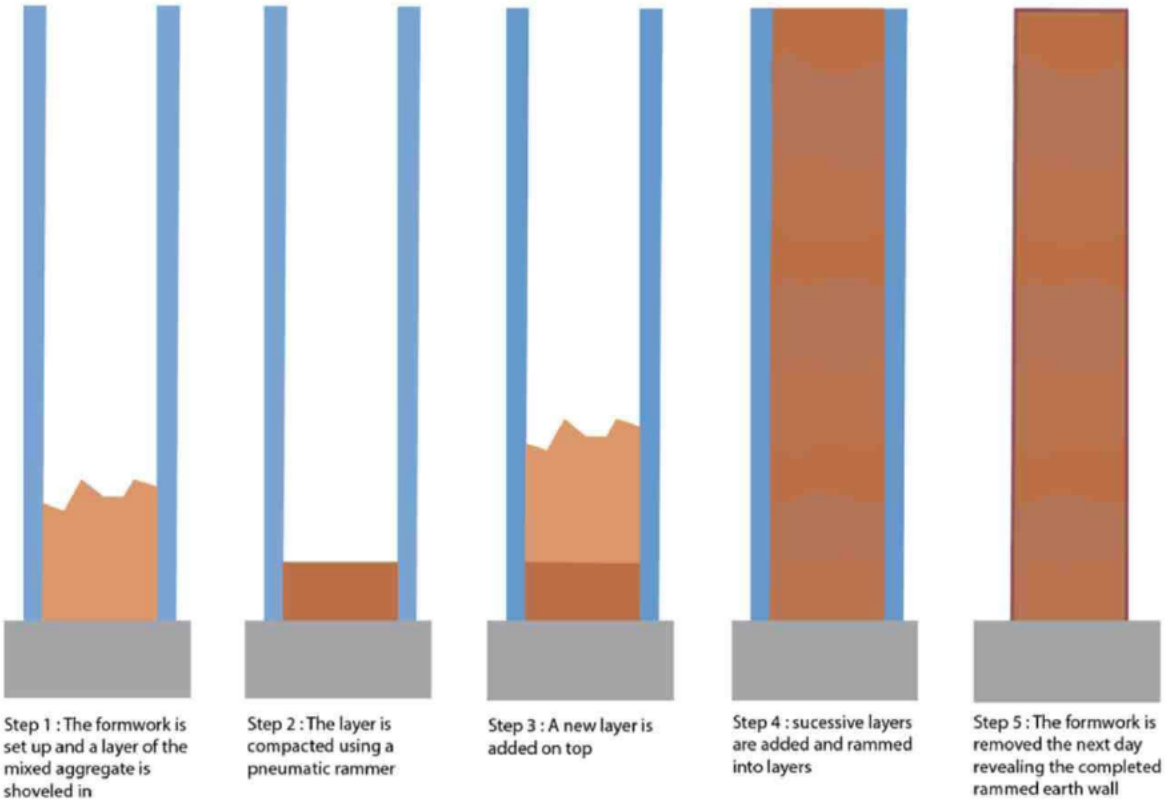


Fig 2.1.1 The process of creating stabilised rammed earth walls.
Adaptation (GreenSpec)



Control-joints allow for the expansion and shrinkage of the walls.

Interlinked SRE panels create the shape of the building's walls

Fig 2.1.2 Port Phillip Estate winery, Victoria, Australia
(Carter)

2.2 History of stabilised earth buildings

Archaeological investigation and excavations have predicted that earth as a building material dates back to approximately 10 000BC, in the neolithic period. It is thought that during this time the growing population put strain on the demand for wood, and the high fire risks caused people to search for alternative building methods (*Hall et al).* Later, stabilisation of earth walls became more common, some of which still stand today. These walls would not have had the luxury of manufactured cement.

The city wall of Nanjing, originally spanning 22 miles, was built 600 years ago. Earth bricks were used to construct the wall, most of which is still standing today, despite centuries of weathering. A key ingredient in the brick's composition was sticky rice, the glutinous grain was used as a form of adhesive for thousands of years across China. The rice was used alongside soil aggregates and small quantities of lime *(Yoon).*

Sundried mud bricks became popular before the invention of concrete for house walls. Also known as adobe, would be a combination of mixing water with earth aggregate similar to that of rammed earth: sand, silt and clay, with organic materials, including manure and straw (Figure 2.3.1) *(Labate).*



Fig 2.3.1 Adobe buildings and ruins of the ancient Pueblo cliff city, Colorado, USA *(Mesa Verde Program)*

2.3 Cement stabiliser

After water, concrete is the most used material in the world due to its low maintenance, high durability properties against environmental factors and external physical forces. The most widely used cement stabiliser in SRE, along with most of the masonry construction industry, is Portland cement. Portland cement accounts for more than 5% of global CO2 emissions making it the largest embodied carbon contributor in the built environment with cement averaging an embodied energy coefficient of 0.74kg_{CO_2e}/kg *(Kupwade-Patil et al.)*.

This requires high levels fossil fuel and energy consumption, further increasing the embodied energy of rammed earth walls, causing the environmental appeal to abate.

For commercialising rammed earth, this has posed many issues as stabilisation of some sort is essential for the properties required by engineers, however many architects/clients are enticed by the more traditional, earth only construction material.

Another negative that steers clients away from the use of cement in rammed earth is the raw material costs can increase. In Australia, buying Portland cement will cost an average of AUD$12 per 20kg bag.

The added expenditure can be heightened in countries/regions where cement resources may not be readily available or must be transported long distances. This inevitably increases the price of cement and the distribution footprint of the LCA.

Although there are many concerns and disadvantages of the use of cement in SRE, there are many counteracting arguments that highlight the benefit of its use.

To begin, the addition of cement reduces the need/quantity of surface treatment or rendering as it creates a high resistance to the environmental elements *(BOSS)*.

Most importantly, adding 5-10% cement drastically improves the durability and strength of the walls.

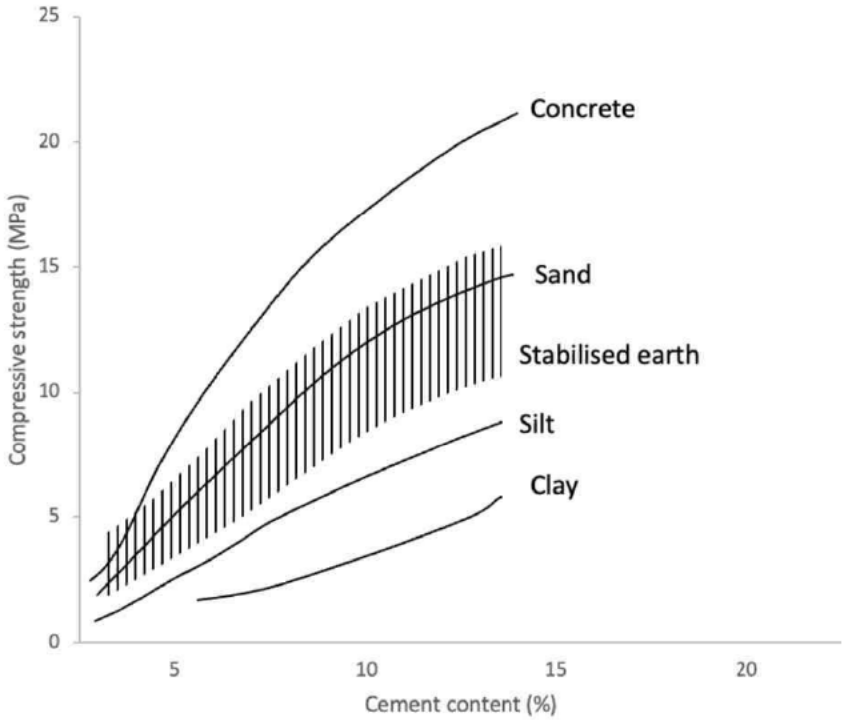


Fig 2.4.1 Relative increase in compressive strength with cement content for different groups of soils *(Hoben and Guillaud)*

3 Materials

To come to an answer for my research question and support my knowledge of differing stabilisers, I've conducted several tests to obtain primary, first-hand data. I have tested and analysed 4 forms of potential stabilisers admixture. The intention is to research alternatives to cement, consequently limiting the amount of cement stabiliser used in stabilised rammed earth (SRE), and further decreasing the environmental impact of the construction industry. I will take into consideration the triple bottom line (TBL); economic, social and environmental impact (in terms of the life cycle analysis LCA) of each material to address the possibilities to attain sustainalblity. Another sample tested was with no added alternative stabiliser, simply with just a small percentage of cement. The four stabilisers I have chosen to explore are pulverised fly ash, wheat dust, blood and bone fertiliser, and animal manure.

3.1 Pulverised fly ash

Pulverised fly ash (PFA) is a siliceous and aluminous material, typically referred to as a Pozzolan. Adoption of PFA in the construction industry has seen a rise with it predominantly being used in the manufacturing of concrete building blocks *(Zakaria et al.)*.

PFA's LCA is reported to be a significant improvement over cement's LCA. As it is the by-product of coal combustion in power station, it is considered an environmental advantage as it re-uses industry waste *(Cement Australia)*. However, many who have a surface level understanding of PFA, perceive it as having a negative impact on the environment as it associated with coal production. Furthermore, PFA has been used to help stabilise concrete for multiple years in a multitude of countries, allowing builders and engineers to trust the structural integrity of PFA in SRE.

In Australia, PFA costs an approximate $9 per 20kg bag which is considerably less expensive than Portland cement, however depending on location, these prices will vary.

3.2 Wheat dust

Wheat dust, commonly referred to as grain dust, is the by-product of when wheat/grain are emptied into silos or storage bins. As the storage vessel is filled, dust is created as the air is displaced *(Agriculture)*. The process of combining wheat and water is most commonly used in conjunction with mixing wheat flour and water.

Currently, wheat dust is in minimal demand and tends to be left to drift off or stored and disposed. Therefore, the adoption of wheat dust in SRE will limit this waste, and with the collection and transport requirements being the sole contributors to the embodied energy, wheat dust has a relatively low LCA. Wheat dust can be highly hazardous and flammable if not managed correctly, thus collecting it, reduces the risk of uncontrolled fires on farms, and subsequently improves the societal influence it would pose on potential clients. The cost for wheat dust would be minor as the product is continuously being produced. There would be expenditure on attaching bins on the side of silos to collect the dust as well as the transport costs from the farm to the site (*Austin)*.

3.3 Blood and bone fertiliser

Blood and bone fertiliser (B&B) was developed as a means to provide an organic source of phosphorus for healthy root development and nitrogen for optimal development of plants *(Tui Blood & Bone)*. The high levels of nitrogen and high coagulation of blood are aspects that allow the B&B to stabilise the rammed earth as it works similar to that of blood clotting. This concept is applied to the B&B being mixed and rammed with aggregates and water.

B&B is a by-product of the freezing works sector; the components are steamed and pulverised finely *(Yardlands)*. The process of grounding the material that make up blood and bone requires the use of automated machinery, as well as the packaging and distributing stages of production. Subsequently increasing the total embodied energy and effecting the LCA of blood and bone fertiliser. The negative connotations that come with using B&B to SRE is knowing that making of the walls requires abattoir waste, therefore, decreasing the appeal, especially for vegan/vegetarian clients. In Australia, the average cost for a 20kg bag is AUD$44, however the price range can be extremely broad from AUD$32.50 *(Martins)* to AUD$62.99 *(Paton)*.

3.4 Animal manure

The manure used for the testing consisted of a combination of horse, wombat, cow and kangaroo, all of which are herbivories, therefore their diet, as well as their manure, is substantially grass/hay.

When manure accumulates as a by-product of livestock systems, it may be regarded as a waste, in which case all system emissions are attributed to the other commodities. However the collection of manure on a mass scale requires the adoption of wheel loaders. In medium to large scale feedlots, wheel loaders are frequently used to remove mounded manure from the enclosure. As the machinery is reasonably large, high levels of fuel consumption will occur when they are in use. The drying process of the manure, if commercially dried, will similarly call for large amounts of energy. Both of these factors impact the LCA of manure in SRE negatively, contradicting the benefits of utilising manure instead of it going to waste. The social impact of using manure in rammed earth walls will affect the market place acceptance as many would be repelled by the aspect. Clients may be sceptical in terms of

aesthetics and smell if the manure had not been sufficiently blended with the mix. Manure is evidently much cheaper than any other substitute being tested. It can be sold for an average of AUD$37/tonne *(GRDC)* which equates to 74c per 20kg.

3.5 TBL comparison of the substitute materials.

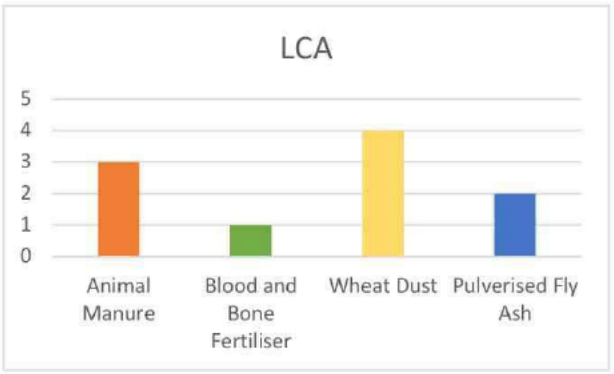


Fig 3.1 LCA comparison of materials

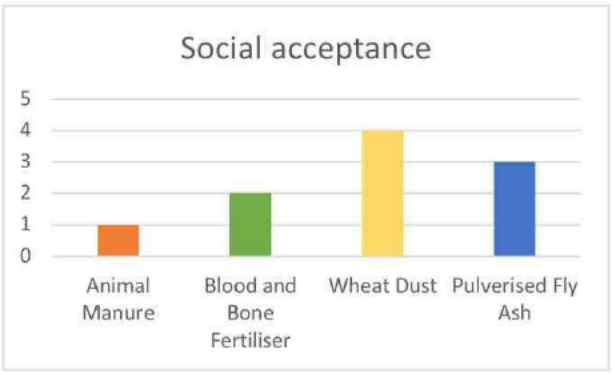


Fig 3.2 Social acceptance comparison of materials

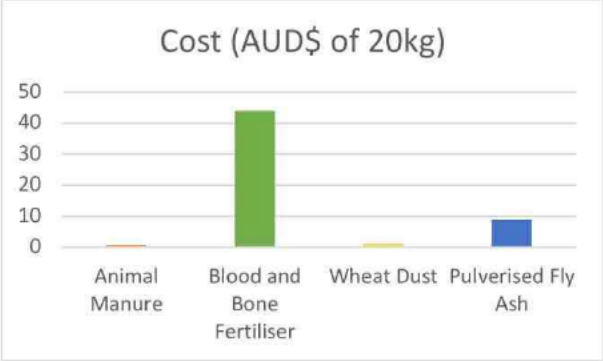


Fig 3.3 Cost comparison of materials

4 Testing Overview and Results

4.1 Sieve analysis

Sieve analysis determines the gradation of an aggregate; distribution of particles by size. To construct rammed earth walls, certain levels of clay, sand, and differing coarseness of rock optimise the characteristics of the road-based-aggregate used. It is the standard specification used by the SRE industry in Australia and other countries. The process involves 13 layers of sieves with varying widths of sieve opening sizes. The sizes that are measured vary from macroscopic granular sizes of less than 0.055mm to greater than 5mm granular material *(Corrosionpedia).*

Method

1. 100 grams of the aggregate is placed in the top pan.
2. Ensuring all pans are secured together, the layers are shaken together manually
3. When it seems that all the aggregate has fallen through the sieves, the pans are taken apart.
4. The amount left in each pan is weighed, and the percentage is retained from this. (figure 4.1.2)



Fig 4.1.1 Sieve Analysis pans



Fig 4.1.1 The road base material separated into sizes

Results

Size (mm)	Percentage retained (%)
5	36.4
2	18.6
1.68	15.3
0.85	8.6
0.6	4.7
0.43	3.3
0.3	2.8
0.21	2.4
0.15	3.2
0.1	3.1
0.075	0.7
0.055	0.5
pan	0.4

Fig 4.1.4 Sieve Analysis results.

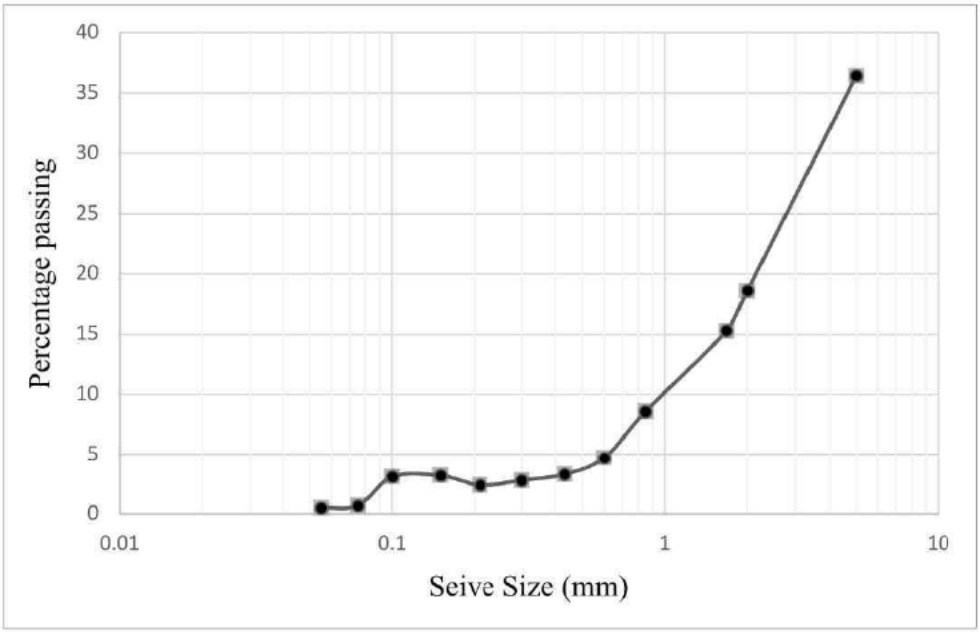


Fig 4.1.5 Graph of sieve Analysis results from road base

I graphed the results from the data I collected *(Fig 4.1.5)*. It is aimed for the curve to simulate a "J shape" as that demonstrates that there is a little bit of every size in the "matrix" *(Lindsay)*. In accordance with Natspec specifications *(Natspec)* (Appendix 1) there is a sought after soil type that correlates nicely with SRE.

Clay and silt material (below .075mm) **< 1.6%**
Sand material (between .075mm and 4.75mm) **≥ 99.1%**
Gravel material (between 4.75mm and 75mm) **> 36.4%**

Fig 4.1.6 Results from primary data sieve analysis

Each of these requirements align with the results from the data we collected *(Fig 4.1.6)*. Therefore, it is concluded that the soil aggregate used for the primary testing accurately displayed the qualities of the exemplary for SRE use.

4.2 How the samples were made

The mix for each sample had equal amounts of cement and substitute material whilst the water and soil aggregate quantities were varied to achieve the optimum consistency (Appendix 2).

Method

1. Each dry material are weighed to precise amounts with the scale being cleaned between each weighing to ensure no added weight was compromising the initial amount. (figure 4.2.1)
2. All dry materials are thoroughly mixed together using hands, all clumps of material must be combined and broken up.
3. Water is weighed and added to the mixed dry material.
4. The water is hand mixed, ensuring all dry material is wet and it is a compactable consistency.
5. The wetted mix is placed into mock-up formwork made of a 90mm PVC stormwater pipe.
6. In equal, handful sized layers, the mix is rammed down using a wood offcut, with the same amount of force for each sample. (figure 4.2.2)
7. The sample is levelled off with finer aggregate being "rammed" on the top using a hammer. (figure 4.2.3)
8. The sample is left to dry for 2 days.
9. The final prototype is removed from the pipe by cutting along the vertical length of the pipe and is then left to cure for 2 weeks.

After 2 weeks of curing, testing was performed on the samples. Rammed earth, at that thickness, takes an approximate on month to completely dry, therefore there may be limitations towards the final results.



Fig 4.2.1 Step 1



Fig 4.2.2 Step 6



Fig 4.2.3 Step 7



Fig 4.2.4 Step 8



Fig 4.2.5 Final result

4.3 MPa testing

MPa (Megapascal) refers to the compressive resistivity of any material by calculating the maximum pressure before cracking of a material (masonry, nylon, steel, etc) occurs. MPa testing is important when observing the characteristics of RE and SRE as it predicts the durability of a wall and accurately portrays the strength resistivity of the final product. Engineers take into the consideration the MPa of the walls to manage the compressive resistance of a wall to retain point loads and general compressive loads. An average SRE wall with a cement percentage of 8-10% has an MPa of 8 – 20mpa in comparison to concreate which measure approximately 20-40Mpa. The MPa testing device used for these samples was

created at our home for the sole purpose of measuring the MPa of rammed earth samples. It is a structure that supports a place for a hydraulic button press and a place for the sample to sit. From this, a pressure gauge was attached where it would read the MPa of the sample.

Method

1. The centre of the hydraulic ram is marked
2. The sample is placed in the centre and a metal plate is placed on top to create an evenly applied load on the sample.
3. The hydraulic press is turned on until failure or cracking occurs in the sample (figure 4.3.2)
4. The sample is removed, and the hydraulic ram is swept clean of all debris and material.



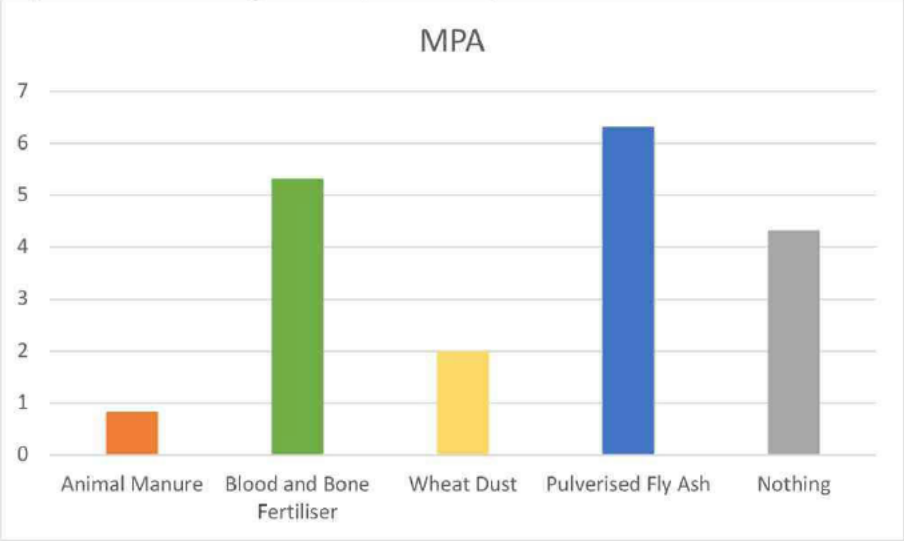
Fig 4.3.1 Pressure gauge reading the MPa of wheat dust



Fig 4.3.2 Step 3

Results

Fig 4.3.3 MPa testing results (MPa units)



(Appendix 3) PFA had the highest MPa recordings averaging 6.33 whilst, on the opposing scale, animal manure scored 0.83MPa. It is suggested both wheat dust and manure decreased the compressive strength value of the samples as their concluding results are considerably less than when there was no added substitute. In comparison, blood and bone fertiliser improved the maximum pressure before cracking by 1MPa and PFA improved it by 2MPa, which in consideration is close to a 50% improvement.

4.4 Density test

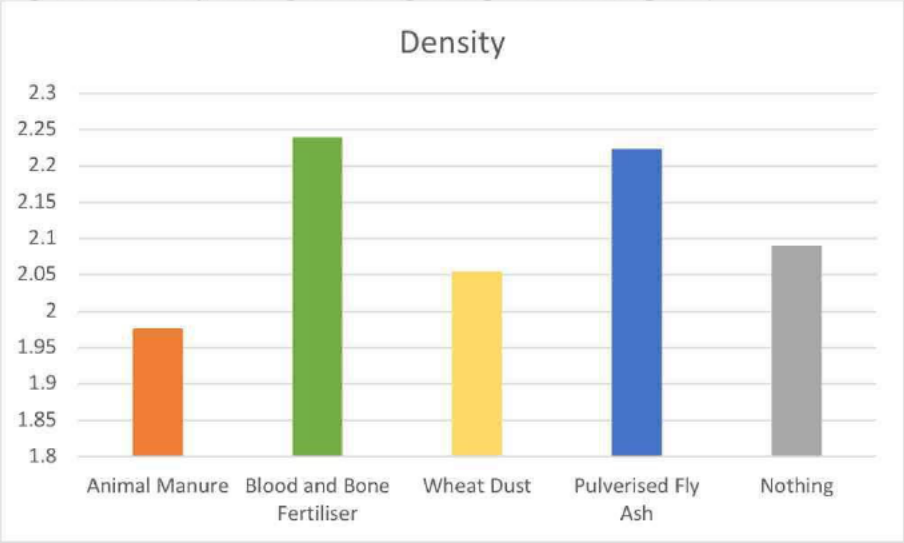
Higher density walls are due to little amounts of voids and porosity, therefore they provide greater strength, resistance and durability to permeability *(Controls Group)*. As rammed earth walls typically have high density, it increases resistivity to hot and cold air from penetrating through the walls. Subsequently, the internal temperature of the building will remain relatively stable *(McCarthy)*. Density is a measured by dividing the total, net weight and dividing it by the total, net volume. Generally, cement SRE walls have an average density of 2250kg/m^3.

Method

1. The internal diameter and height of the PVC pipe was measured
2. Volume was calculated
3. Each sample was weighed, and the mass was recorded
4. The mass was divided by the volume to find the density of each sample. (Appendix 4)

Results

Fig 4.4.1 Density testing results (grams per millilitre g/mL)



(Appendix 5) Once again, animal manure and wheat dust, diminished the value of the material. Represented in figure 4.4.1, it is shown that manure decreases the density levels a relatively small amount (around 0.1g/mL). However, blood and bone fertiliser contrasts this by improving the density by an average 0.21g/mL. PFA was the next best in terms of density, considering PFA, on its own, is an extremely light weight material this is a notable material for increasing the samples density.

4.5 Water absorption test

Water absorption testing determines the weatherability of exterior wall when exposed to recurring rain events. SRE is recognised for its longevity, therefore, erosion and cracks due to inadequate water resistivity will jeopardize the durability. Testing how absorbent a wall is helps establish the percentage of Plasticure required as a waterproof additive. It also helps determine the application rate of acrylic surface binders to prevent any internal wall dusting after the walls have completely dried.

Method

1. The largest section of each sample that remained after MPa testing is kept to test the water absorption levels. (figure 4.5.1)
2. The samples are spaced out and laid in a deep baking tray
3. The baking tray is filled with 60mm of water
4. The samples are left for 24 hours to absorb the water.
5. The samples are broken in half using a chisel and mallet (figure 4.5.2)
6. The depth of which the water soaked into the substrate of the sample is recorded. (figure 4.5.3)



Fig 4.5.1 Step 1



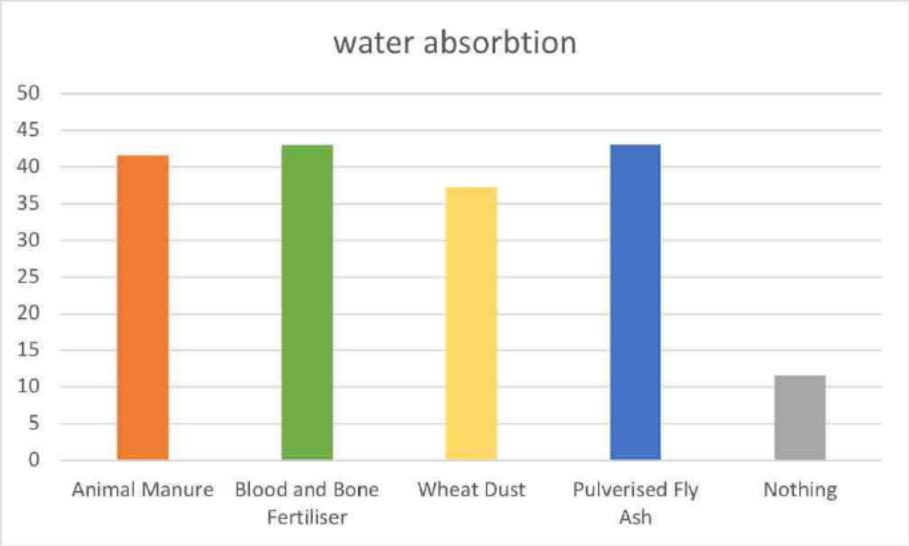
Fig 4.5.2 Step 5



Fig 4.5.3 Step 6

Result

Fig 4.5.4 Water absorption testing results (mm absorbed into sample)



(Appendix 6) This testing depicted that additive stabiliser negatively comprises the water absorption of SRE walls. The samples illustrates how the water absorbed all the way through to the core whilst with no added stabiliser, the water only reach a maximum of 14mm into the sample. Wheat dust proved to be most effective out of the added material samples however it was nowhere near as water resistant. This diminishes the durability and longevity of these material if no added plasticure or acrylic is added.

5 Analysis

For the 3 primary tests that were conducted I have ranked them 1-4 (4 being the best, 1 being the worst) to calculate which material would be the most appropriate for stabilising rammed earth.

Fig 5.1 Material ranking

	Density (g/mL) 2/1/24	MPA	Water absorption (mm)	Average out of 4	**Ranking**
PFA	3	4	2	3	**first**
Wheat Dust	2	2	4	2.6	**second**
B&B	4	3	2	3	**first**
Manure	1	1	3	1.6	**third**

Figure 5.1 displays PFA and blood and bone being the optimal selection however the importance and relevance of each test plays a large role in the selection of the most appropriate material. When considering water absorption, a higher percentage of plasticure and acrylic can be utilised to prevent mould and water effecting the quality of the walls, however it is not preferable. On the other hand, a weakened MPa will prevent the walls from being able to be approved by engineers or it will result in increased risk of crumbling/cracking walls in the building. On top of this, these are ordinal rankings and the difference between each rank can vary drastically. For this, I believe PFA has a higher importance than blood and bone as the difference in density is minor. To further this, the MPa of a wall is more relevant than density and adjusting the density is simpler in contrast to the MPa.

Fig 5.2 Material ranking with LCA

	Density (g/mL) 2/1/24	MPA	Water absorption (mm)	LCA	Average out of 4	**Ranking**
PFA	3	4	2	2	2.75	**second**
Wheat Dust	2	2	4	4	3	**first**
B&B	4	3	2	1	2.5	**third**
Manure	1	1	3	3	2	**fourth**

However, the function in terms of structural integrity, durability and water resistivity are not the sole depicters of the most effective stabiliser. There is a constant, industry wide debate that challenges environmental costs vs functionality costs vs financial costs.

When analysing the LCA of each product, each material has a relatively low impact as they are a by-product of agricultural and industrial waste. Blood and Bone fertiliser was marked as having the greatest cost to the environment as the material requires processing. The collection process of animal manure can be extravagant however the drying procedure of the manure is what decreases the environmental sustainability of the stabiliser, however if airdryed outside, time would be the only negative barrier.

Pulverised fly ash and wheat dust both require no added processing after it has been created. However, PFA is still packaged for mass sales therefore, if not disposed of correctly, may correspond with decreased environmental sustainability. In comparison, wheat dust will come straight from the farm as currently it is unable to be purchased from a distributer.

The addition of LCA to the ordinal rankings has relatively adjusted the final results. Blood and bone fertiliser still remains the least favourable option however wheat dust has advanced to the top ranking. PFA is only 0.25 behind wheat dust, suggesting that between the two tables, it is the most preferred.

6 Conclusion

At the start of this research essay I was excited at the prospect of using animal and plant based proteins, or any source of environmentally sensible and available stabilisers as a way to stiffen or harden rammed earth substrate, however my testing has proven this theory unsuccessful.

The conclusion, after examining the test results, must be that cement remains the most effective stabiliser for rammed earth. However pulverised fly ash adds further strengthening and malleability to the walls seen by the improvements of 2Mpa to the samples with only cement. Moreover, PFA's density averaged out to 2.22 g/mL, coming a close second to Blood and Bone in terms of ranking. When looking at the sustainability aspect of PFA, it's good in the short term as it is considered a by-product, however once coal reduction ceases then another alternative must be discovered.

7 Appendix

A.1 Sieve analysis Natspec specifications *(Natspec)*

Clay and silt material (below .075mm) **< 20%**
Sand material (between .075mm and 4.75mm) **≥ 50%**
Gravel material (between 4.75mm and 75mm) **> 30%**

A.2 Quantity of each material in samples

	earth amount (grams)	cement amount (grams)		water amount (grams) based upon consistency		Other amount (grams)	
Pulverised Fly Ash	2000	5%	100	10.50%	210	5%	100
	2000	5%	100	10.50%	210	5%	100
	2000	5%	100	10.50%	210	5%	100
Wheat Dust	2000	5%	100	12.50%	250	5%	100
	2000	5%	100	12.50%	250	5%	100
	2000	5%	100	12.50%	250	5%	100
Blood and Bone Fertiliser	2000	5%	100	10%	200	5%	110
	2000	5%	100	10%	200	5%	110
	2000	5%	100	10%	200	5%	110
Nothing	2000	5%	100	11%	220	0%	0
	2000	5%	100	11%	220	0%	0
	2000	5%	100	11%	220	0%	0
Animal Manure	2000	5%	100	17.50%	350	5%	100
	2000	5%	100	17.50%	350	5%	100
	2000	5%	100	17.50%	350	5%	100

A.3 MPa testing results (MPa units)

Pulverised Fly Ash	6
	7
	6
average	6.33

Wheat Dust	2
	2
	2
average	2

Animal Manure	1
	1
	0.5
average	0.83

Blood and Bone Fertiliser	6
	5
	5
average	5.33

Nothing	4
	4
	3
average	4.33

A.4 Density calculations

$$\textbf{Volume} = \pi \times \text{radius}^2 \times \text{height}$$

$$1074.85\text{mL}^3 = \pi 4.3\text{cm}^2 \times 18.5\text{cm}$$

$$\textbf{Density} = \frac{\text{mass}}{\text{volume}}$$

$$\text{Density} = \frac{\text{mass}}{1074.85\text{mL}^3}$$

A.5 Density testing results (grams per millilitre g/mL)

Pulverised Fly Ash	2.217053542
	2.236591152
	2.217983905
average	2.223873333

Wheat Dust	2.0700563
	2.0765688
	2.017956
average	2.0548533

Animal Manure	1.9612039
	1.9946969
	1.9723682
average	1.9760833

Blood and Bone Fertiliser	2.230078616
	2.248685863
	2.239382239
average	2.239376667

Nothing	2.0886635
	2.0933153
	2.0895939
average	2.09052

A.6 Water absorption testing results (mm absorbed into sample)

Pulverised Fly Ash	total
	total
	total
average	43

Wheat Dust	35
	34
	total
average	37.3

Animal Manure	total
	total
	39
average	41.6

Blood and Bone Fertiliser	total
	total
	total
average	43

Nothing	9
	12
	14
average	11.6

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